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$$\rho^{-1}(\alpha(s) - c) = \rho^{-1}(\rho(\bar{\alpha}(s))) = \bar{\alpha}(s) \quad \cdots\cdots (65)$$

Finally, ρ^{-1} is linear so that

$$\rho^{-1}(\alpha(s) - c) = \rho^{-1}(\alpha(s)) - \rho^{-1}(c) \quad \cdots\cdots (66)$$

Comparing (65) and (66), one obtains,

$$\bar{\alpha}(s) = \rho^{-1}(\alpha(s)) - \rho^{-1}(c) \quad \cdots\cdots (67)$$

(63) reveals the fact that the 2 curves α and $\bar{\alpha}$ are related by a rigid motion

- We will prove the existence part of the Fundamental theorem in the next class.

Lecture 3

Choose $s_0 \in I$, a point $p_0 \in \mathbb{R}^3$, and a positively oriented orthonormal initial frame $\{T_0, N_0, B_0\}$ with $F_0 = [T_0, N_0, B_0]$ being an orthogonal matrix of unit determinant. We now have to construct a curve from prescribed $k(s) > 0$ and $\tau(s)$.

Define
$$A(s) = \begin{bmatrix} 0 & -k(s) & 0 \\ k(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{bmatrix} \quad \cdots\cdots (1)$$

Check that $A(s)^T = -A(s)$

Solve the matrix differential equation $F'(s) = F(s)A(s)$ with $F(s_0) = F_0$ $\quad \cdots (2)$

$$\begin{bmatrix} T_1'(s) & N_1'(s) & B_1'(s) \\ T_2'(s) & N_2'(s) & B_2'(s) \\ T_3'(s) & N_3'(s) & B_3'(s) \end{bmatrix} = \begin{bmatrix} T_1(s) & N_1(s) & B_1(s) \\ T_2(s) & N_2(s) & B_2(s) \\ T_3(s) & N_3(s) & B_3(s) \end{bmatrix} \begin{bmatrix} 0 & -k(s) & 0 \\ k(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{bmatrix} \quad \cdots (3)$$

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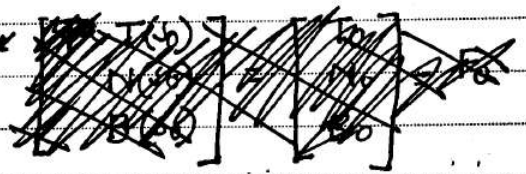
Explicitly

$$\begin{bmatrix} T_1'(s) & N_1'(s) & B_1'(s) \\ T_2'(s) & N_2'(s) & B_2'(s) \\ T_3'(s) & N_3'(s) & B_3'(s) \end{bmatrix} = \begin{bmatrix} k(s)N_1(s) & -k(s)T_1(s) - \tau(s)B_1(s) & \tau(s)N_1(s) \\ k(s)N_2(s) & -k(s)T_2(s) - \tau(s)B_2(s) & \tau(s)N_2(s) \\ k(s)N_3(s) & -k(s)T_3(s) - \tau(s)B_3(s) & \tau(s)N_3(s) \end{bmatrix}$$

By equating the columns, one has

$$\left.\begin{aligned} T'(s) &= k(s)N(s) \\ N'(s) &= -k(s)T(s) - \tau(s)B(s) \\ B'(s) &= \tau(s)N(s) \end{aligned}\right\} \cdots (4)$$

In other words, we are looking for the unique solution of the linear differential system with continuous coefficients with the initial value $F(s_0) = [T(s_0), N(s_0), B(s_0)] = [T_0, N_0, B_0] = F_0$



Now, $\frac{d}{ds}\left(F(s)F(s)^T\right) = F'(s)F(s)^T + F(s)\left(F'(s)\right)^T$

$= F'(s)A(s)F(s)^T + F(s)A^T(s)F(s)^T$; [By (2)]

$= F(s)\left(A(s) + A^T(s)\right)F(s)^T$

$= 0$; $[\because A^T(s) = -A(s)]$

Thus $F(s)F(s)^T = F_0(s)F_0(s)^T = I_3$ ..... (5)

$\therefore F^{-1}(s) = F(s)^T$. Hence $F(s)F(s)^T = F(s)^T F(s) = I_3$. Since $F(s)$ is an orthogonal matrix, its columns are orthonormal.

Now the function $g(s) = \det F(s)$ ..... (6)

is continuous because the entries of $F(s)$ are continuous and the determinant is a polynomial in those entries.

We know $g(s) \in \{-1, 1\}$ for every $s \in I$, $g(s_0) = 1$

If $g(s) = -1$ anywhere in I, the intermediate value theorem on the interval between s_0 and s_1 would give

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some s_* with $g(s_*) = 0$,

a contradiction.

Hence, $\boxed{\det F(s) = 1 \quad \forall s \in I}$ ..... (7)

- Note that $s_0 \in I$ is just a prescribed parameter value at which the initial data $F(s_0)$ is given. It does it mean that the parameter s is supposed to be always greater or equal to s_0.

Now define $\alpha(s) = p_0 + \int_{s_0}^{s} T(u)\, du$ ..... (8)

Then by the fundamental theorem of calculus,

$$\alpha'(s) = T(s) \quad \text{..... (9)}$$

And since $\{T(s), N(s), B(s)\}$ constitutes an orthonormal basis at each parameter value $s \in I$,

$$|\alpha'(s)| = |T(s)| = 1 \quad \text{..... (10)}$$

Thus the curve $\alpha(s)$ defined by (8) is arc-length parametrized and hence regular.

Furthermore, $\alpha''(s) = T'(s)$

$= k(s) N(s)$; [By (4)] ..... (11)

Since $|N(s)| = 1$ by orthonormality of the basis $\{T(s), N(s), B(s)\}$ and $k(s) > 0$, $\forall s \in I$, one obtains

$$|\alpha''(s)| = k(s), \quad \frac{\alpha''(s)}{|\alpha''(s)|} = N(s) \quad \text{..... (12)}$$

Hence, the curvature of $\alpha(s)$ is indeed prescribed by $k(s)$ and its principal normal is the constructed vector $N(s)$. Now, since $\{T(s), N(s), B(s)\}$ constitutes an orthonormal system and $\det F(s) = 1$, one must have

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$$\vec{B}(s) = \vec{T}(s) \wedge \vec{N}(s) \quad \cdots\cdots\cdots (13)$$

So, the binormal of the curve $\alpha(s)$ is the constructed vector $B(s)$.

Finally, by (4), we have $B'(s) = \tau(s) N(s)$,

So the torsion of $\alpha(s)$ is precisely the prescribed function $\tau(s)$.

This proves existence and completes the fundamental theorem.

QED

Curve	Base point at $s=0$	Frenet tripod
original curve α	$p_0 = \alpha(0)$	$\{T_0, N_0, B_0\}$
Normalized curve $\beta = M\alpha$	$O = \beta(0)$	$\{e_1, e_2, e_3\}$

"End of lecture 3"

Lecture 4

The local canonical form

The key idea is to fix the Frenet tripod at one point and use it as a coordinate system for the nearby curve.

Let $\alpha: I \to \mathbb{R}^3$ be a curve parametrized by arc-length without singular points of order 1. k is a differentiable function with $k(s_0) > 0$. By continuity of k, we have $k > 0$ in a nbhd of s_0, so that the Frenet frame is defined there. Put $h = s - s_0$ and write $\tilde{\alpha}(h) = \alpha(s_0 + h) \quad \cdots\cdots (1)$

Then $\tilde{\alpha}'(h) = \alpha'(s_0+h) \frac{d}{dh}(s_0+h) = \alpha'(s_0+h) \quad \cdots\cdots (2)$

Thus $|\tilde{\alpha}'(h)| = 1$; [$\because |\alpha'(s)| = 1$ as α is arc-parametrized]

Write $p_0 = \alpha(0)$ (we dropped the tilde and we'll be denoting s by h from now on)

$T_0 = T(0)$, $N_0 = N(0)$, $B_0 = B(0) \quad \cdots\cdots\cdots (3)$

We place the spatial origin at p_0 and use T_0, N_0, B_0 as the positive x, y, z directions, resp.